

MTH204 (Section A)
Graded Worksheet–3

Name:

Roll No.:

Important Notice: The use of electronic gadgets is strictly prohibited during the tutorial. This includes mobile phones, tablets, or any device containing class notes, but anything handwritten is completely allowed.

1. If $y(x) = x$ is a solution of the D.E.

$$y'' - \left(\frac{2}{x^2} + \frac{1}{x} \right) (xy' - y) = 0,$$

then find its general solution.

2. Solve the I.V.P.

$$6y'' - y' - y = 0, \quad y(0) = -0.5, \quad y'(0) = 1.25.$$

3. Deduce whether the following functions are linearly dependent or linearly independent. Show the details of your work.

(a) $x^2, x^2 \ln x$ ($x > 1$)

(b) $\sin 2x, \cos x \sin x$ ($x < 0$)

4. If $y(x)$ is the solution of the I.V.P.

$$y'' + 4y' + 4y = 0, \quad y(0) = 0, \quad y'(0) = 2,$$

then the value of $y(\log 2)$ is _____.

5. Find the solution of the I.V.P.

$$y'' + y' + 2y = 0, \quad y(0) = 0, \quad y'(0) = 2.$$

6. Let $y(x)$ be the solution of the following initial value problem:

$$x^2 \frac{d^2 y}{dx^2} - 4x \frac{dy}{dx} + 6y = 0, \quad x > 0,$$

$$y(2) = 0, \quad \frac{dy}{dx}(2) = 4.$$

Then $y(4) =$ _____.

Solution:

Solution:

Solution:

Solution: